

A counterexample to Rosenfeld’s hypergraph generalization of Vizing’s theorem

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Machine-generated note

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Abstract

Rosenfeld conjectured (Open Problem Garden, 2007) that every simple d -uniform hypergraph in which each $(d-1)$ -set lies in at most r edges has an $(r+d-1)$ -edge-colouring in which any two edges sharing $d-1$ vertices receive distinct colours; for $d=2$ this is Vizing’s theorem. We show that the conjecture fails for $d=100$, $r=200$: a probabilistic construction, whose main ingredient is an entropy bound on the number of large matchings in a regular linear hypergraph, yields a simple 100-uniform hypergraph with maximum 99-codegree at most 200 that needs at least 451 colours, against the conjectured 299. The same method gives, for every $d \geq 15$, hypergraphs with maximum codegree $2d$ needing $(1-o(1))d^2/(5 \log d)$ colours, so the conjectured bound $r+d-1$ is off by an unbounded factor. The counterexample is non-constructive and astronomically large: the proof produces it only for $n \geq 10^{456}$ vertices per class, i.e. about 10^{458} vertices and 10^{45000} edges. It cannot be instantiated or spot-checked, and the disproof rests entirely on the correctness of the counting lemma, every ingredient of which was verified separately by the referee on small instances.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

A hypergraph H is *d-uniform* if every edge has exactly d vertices, and *simple* if it has no repeated edges (so $E(H) \subseteq \binom{V(H)}{d}$). A *facet* of an edge e is one of its d subsets of size $d-1$; the *codegree* of a $(d-1)$ -set is the number of edges containing it. A *facet colouring* of H is a map $E(H) \rightarrow C$ such that two distinct edges sharing a facet, i.e. sharing $d-1$ vertices, receive different colours, and $\chi_f(H)$ denotes the least number of colours in a facet colouring. All logarithms are natural.

The problem posed by Rosenfeld on the Open Problem Garden [1] reads, verbatim:

Conjecture 1.1 (Rosenfeld [1]). Let H be a simple d -uniform hypergraph, and assume that every set of $d-1$ points is contained in at most r edges. Then there exists an $r+d-1$ -edge-coloring so that any two edges which share $d-1$ vertices have distinct colors.

In the notation above the conjecture states $\chi_f(H) \leq r+d-1$ whenever every $(d-1)$ -set has codegree at most r . For $d=2$ the hypothesis says $\Delta(H) \leq r$ and the conclusion is $\chi'(H) \leq r+1$, which is Vizing’s theorem [2]. The OPG entry remarks that Kempe-chain arguments, the main tool for Vizing’s theorem, do not seem to work for $d \geq 3$; apart from a 2009 comment asking for a reference, the entry carries no further information, and the referee’s literature search ([Appendix B](#)) found no paper attacking the conjecture.

The facet hypergraph. It is convenient to translate the problem. Given H , let $F = F(H)$ be the hypergraph whose vertices are the $(d-1)$ -subsets of $V(H)$ that are facets of some edge, with one edge $\{ \text{the } d \text{ facets of } e \}$ for each $e \in E(H)$. Then F is d -uniform, its maximum degree is the maximum codegree of H , and F is *linear* (two distinct edges share at most one vertex), because two distinct d -sets cannot have two common $(d-1)$ -subsets. A facet colouring of H is exactly a proper edge colouring of F , so [Conjecture 1.1](#) asserts $\chi'(F) \leq \Delta(F) + d - 1$ for every linear d -uniform F that is *realisable* as a facet hypergraph. Realisability is essential: for general linear hypergraphs the bound is false by a wide margin (a projective plane of order q is a linear $(q+1)$ -uniform hypergraph with $\Delta = q+1$ and $\chi' = q^2 + q + 1$), but a projective plane is not a facet hypergraph, since d -sets pairwise meeting in $d-1$ points form a star or lie inside a $(d+1)$ -set. The construction below respects this constraint by building an actual H .

Related work. The conjecture is sharp: the referee computed exact facet chromatic numbers of complete d -uniform hypergraphs $\binom{[d+k]}{d}$ (here $r = k+1$ and $r+d-1 = d+k$) and found the bound attained with equality at $(d, d+k) = (2, 3), (2, 5), (3, 4), (3, 5), (3, 6), (4, 5), (5, 6), (5, 7), (6, 7), (7, 8)$, and never exceeded; 2357 random instances with $d \in \{3, 4, 5\}$ on at most $d+4$ vertices gave no violation either. Any counterexample therefore has to be large and sparse, which is what the probabilistic construction produces.

Several neighbouring lines of work concern edge colourings of linear hypergraphs in other regimes and are not in conflict with our result. Pippenger and Spencer [3] and Kahn [4] proved $\chi'(F) = (1 + o(1))\Delta(F)$ for linear k -uniform F with k fixed and $\Delta \rightarrow \infty$; our hypergraphs have $k = d$ and $\Delta \leq 2d$, so k is of the order of Δ and these theorems do not apply (and projective planes show they cannot be extended to that regime). Alon and Kim [5] study t -simple uniform hypergraphs, again for fixed uniformity and large degree. The Berge–Füredi conjecture [6, 7], $\chi'(F) \leq \Delta([F]_2) + 1$ for linear F , bounds the chromatic index by the maximum degree of the *shadow graph* rather than of F , and has seen recent progress [8, 9, 10]; Kang, Kelly, Kühn, Methuku and Osthus [11] solved a problem of Erdős on the (ordinary) chromatic index of hypergraphs with bounded codegree. None of these addresses colourings in which only edges sharing $d-1$ vertices must differ. To the best of the referee’s search, the counterexample below is new; see [Remark 4.5](#).

The result. Fix $d = 100$ and let n be a multiple of 5. Take disjoint classes $V_1, \dots, V_{100}$ of size n , let H_0 be the complete 100-partite 100-uniform hypergraph (all transversal 100-sets), retain each edge independently with probability $100/n$, and delete every retained edge containing a facet of codegree above 200. The resulting hypergraph H is simple, 100-uniform, and has maximum 99-codegree at most 200. A first-moment count of large matchings in the facet hypergraph shows that, with overwhelming probability, no colour class can contain $n^{99}/5$ edges, while H keeps more than $90 n^{99}$ edges; hence $\chi_f(H) \geq 451$.

Theorem 1.2 (Main theorem). *Let $n \geq 10^{456}$ be a multiple of 5. There exists a simple, 100-partite, 100-uniform hypergraph H with $100n$ vertices such that*

1. *every 99-set of vertices is contained in at most 200 edges, and*
2. *every facet colouring of H uses at least 451 colours.*

Corollary 1.3. *[Conjecture 1.1](#) is false for $d = 100$ and $r = 200$: the conjectured bound is $r + d - 1 = 299 < 451 \leq \chi_f(H)$.*

Proof. Immediate from [Theorem 1.2](#). □

Idea of proof. Let $T = n^{99}$. The facet hypergraph F of H_0 has $N = 100T$ vertices, is 100-uniform, n -regular and linear, and a set of edges of H_0 is a legal colour class iff the corresponding edges of F form a matching. [Lemma 2.3](#) bounds the number $a_m(F)$ of m -matchings of a regular linear hypergraph from above; for $m = T/5$ and sampling probability $p = 100/n$ it gives $\mathbb{E}[\#m\text{-matchings in the sampled } F] \leq$

$e^{-T/5}$ (Proposition 3.1), so with probability at least $1 - e^{-T/5}$ every colour class of the sampled hypergraph H_{rand} , and of every subhypergraph, has fewer than $T/5$ edges. Pruning the edges through overloaded facets removes only an expected $100(e/4)^{100}$ fraction of the edges (Markov's inequality applied to 2^X , $X \sim \text{Bin}(n-1, 100/n)$), so some outcome has both properties and more than $90T$ edges; 450 colours would cover fewer than $450 \cdot T/5 = 90T$ edges (Proposition 3.2). [full proof]

The exponent in the first-moment count improves as d grows, and the same argument shows that the conjectured bound is wrong by an unbounded factor.

Theorem 1.4 (Large uniformity). *For every integer $d \geq 15$ there exists a simple d -uniform hypergraph H_d in which every $(d-1)$ -set lies in at most $2d$ edges and*

$$\chi_f(H_d) \geq (1 - 2d(e/4)^d) \frac{d^2}{\lceil 5 \log d \rceil}.$$

Consequently $\chi_f(H_d)/(r+d-1) \geq (1 - o(1)) d/(15 \log d) \rightarrow \infty$ as $d \rightarrow \infty$, where $r = 2d$.

Idea of proof. Repeat the construction with d classes of size n , sampling probability d/n , pruning threshold $2d$ and matching size xT with $x = \lceil 5 \log d \rceil/d$, $T = n^{d-1}$. For fixed $d \geq 15$ the first-moment exponent is negative, so letting $n \rightarrow \infty$ (through multiples of d) gives an outcome with at least $dT(1 - 2d(e/4)^d)$ edges and no colour class of size xT . The limits are taken in the order “fix d , then $n \rightarrow \infty$ ”; no uniformity in d is claimed or needed. [full proof]

2 Counting matchings in regular linear hypergraphs

The counting lemma is proved by the entropy method, in the manner of Radhakrishnan's proof of Brégman's theorem [12, 13] and of the Kahn–Lovász theorem [14, 15]: a uniformly random perfect matching is revealed vertex by vertex in a random order, and the entropy of each reveal is bounded by the logarithm of the total weight of the still-available edges. The hypergraph version needs two modifications: edge weights, and a small allowance ε for edges that meet a matching edge in two or more vertices.

Lemma 2.1 (Weighted perfect matchings). *Let $s \geq 2$ and let K be an s -uniform hypergraph on L vertices with positive edge weights w , such that every vertex has weighted degree $\sum_{e \ni v} w(e) \leq \Delta$. Put $Z = \sum_M \prod_{e \in M} w(e)$, the sum over perfect matchings M of K . Call an edge e bad relative to M if $|e \cap f| \geq 2$ for some $f \in M$; note that every edge of M is bad relative to M . Suppose that for every perfect matching M and every vertex v ,*

$$\sum_{e \ni v, e \text{ bad relative to } M} w(e) \leq \varepsilon \Delta. \quad (1)$$

Then

$$\log Z \leq \frac{L}{s} \left[\log \Delta + \int_0^1 \log(u^{s-1} + \varepsilon) du \right]. \quad (2)$$

Idea of proof. Let M be a perfect matching drawn with probability $\prod_{e \in M} w(e)/Z$, so that $\log Z = H(M) + \mathbb{E} \sum_{e \in M} \log w(e)$. Give the vertices independent uniform labels in $[0, 1]$ and reveal $M(v)$, the matching edge at v , in decreasing order of labels. By the chain rule and Gibbs' inequality, $\log Z \leq \sum_v \mathbb{E}[\mathbf{1}\{v \text{ has the largest label in } M(v)\} \log S_v]$, where S_v is the total weight of the edges at v disjoint from all previously revealed matching edges. Given the label t of v , the event has probability t^{s-1} , and a non-bad edge at v is still available with probability $t^{s(s-1)}$ because its other $s-1$ vertices lie in $s-1$ distinct matching edges other than $M(v)$. Jensen's inequality and the substitution $u = t^s$ give (2). [full proof]

As $\varepsilon \rightarrow 0$ the integral in (2) tends to $\int_0^1 (s-1) \log u du = -(s-1)$, which is the form in which the original writeup used the lemma. The rate matters, and we make it explicit.

Lemma 2.2 (Quantitative form of the integral). *For every integer $s \geq 3$ and every $\varepsilon \in (0, 1)$,*

$$\int_0^1 \log(u^{s-1} + \varepsilon) du \leq -(s-1) + (s+1) \varepsilon^{1/(s-1)}.$$

Repair. The referee’s only substantive objection was that the writeup asserts $\int_0^1 \log(u^{s-1} + \varepsilon) du = -(s-1) + o(1)$ without a rate, whereas the true correction is $\approx (s-1)\varepsilon^{1/(s-1)}$, a 99th root for $s = 100$, and it is multiplied by $L/s \approx 80 n^{99}$ against a margin of only $0.2 n^{99}$ in [Proposition 3.1](#). [Lemma 2.2](#) is the missing quantification; its proof splits the integral at $u_0 = \varepsilon^{1/(s-1)}$ and uses $\log(1+y) \leq y$ above u_0 . The constant $s+1$ is not optimal ($s-1$ is the asymptotically correct one) but suffices.

Idea of proof. Below $u_0 = \varepsilon^{1/(s-1)}$ the integrand is at most $\log(2\varepsilon)$; above u_0 it is at most $(s-1) \log u + \varepsilon u^{-(s-1)}$, and $\int_{u_0}^1 \varepsilon u^{-(s-1)} du \leq u_0/(s-2)$. The $\log \varepsilon$ terms cancel exactly. [\[full proof\]](#)

Lemma 2.3 (Matchings in a regular linear hypergraph). *Let $s \geq 3$ and $x \in (0, 1)$. Let F be an s -uniform, D -regular, linear hypergraph on N vertices such that $m = xN/s$ is an integer, and put $u = N - sm = (1-x)N$ and $b = (s-1)u$. Assume $b \geq 4$ and*

$$\varepsilon := \frac{x}{D} + \frac{s^3}{b} < 1.$$

Then the number $a_m(F)$ of matchings of size m in F satisfies

$$\log a_m(F) \leq m(\log(D/x) - (s-1)) - N(1-x)\log(1-x) + (m+u)(s+1)\varepsilon^{1/(s-1)}. \quad (3)$$

Repair. The writeup states (3) with an unquantified $o(N)$ in place of the last term, arising from three sources: the $o(1)$ of the integral, an $O(N^{-1})$ in ε from bad auxiliary edges, and Stirling’s formula. Here all three are explicit: the integral by [Lemma 2.2](#), the bad-edge fraction by a direct count giving s^3/b , and Stirling is replaced by the one-sided bounds $\log b! \geq b \log b - b$ and $\binom{b}{s-1} \leq b^{s-1}/(s-1)!$, which are all that the argument needs and carry no error term. The referee verified the auxiliary construction, the weighted regularity and the identity (4) exactly on five small instances ([Appendix B](#)).

Idea of proof. Adjoin a set B of b new vertices and, for every old vertex a and every $C \in \binom{B}{s-1}$, an auxiliary edge $\{a\} \cup C$ of weight $w = \frac{(1-x)D/x}{\binom{b}{s-1}}$; the original edges keep weight 1. The resulting K is weighted-regular of degree $\Delta = D/x$, and its perfect matchings correspond to pairs (m -matching of F , partition of B into $(s-1)$ -blocks assigned to the u uncovered old vertices), so

$$Z = a_m(F) \cdot \frac{b!}{((s-1)!)^u} \cdot w^u. \quad (4)$$

Linearity of F and the shape of the auxiliary edges make hypothesis (1) hold with the stated ε . [Lemmas 2.1](#) and [2.2](#) bound $\log Z$ from above, the one-sided Stirling bounds bound the completion factor in (4) from below, and subtraction gives (3). [\[full proof\]](#)

3 The random hypergraph

Throughout this section $d = s = 100$, n is a multiple of 5, $T = n^{99}$, $V_1, \dots, V_{100}$ are disjoint classes of size n , and H_0 is the hypergraph of all transversal 100-sets, so $|E(H_0)| = n^{100}$. Its facet hypergraph F has as vertices the $100 n^{99} = 100T$ transversal 99-sets; each lies in exactly n edges of H_0 , so F is 100-uniform, n -regular and linear on $N = 100T$ vertices. Retain each edge of H_0 independently with probability $p = 100/n$; call the result H_{rand} . A set of edges of H_{rand} is a legal colour class iff the corresponding edges of F form a matching.

Proposition 3.1 (No large colour class). *Let $m = T/5$ and let A be the event that H_{rand} contains no m edges that are pairwise facet-disjoint. If $n \geq 10^{456}$ then $\Pr(A^c) \leq e^{-T/5}$.*

Repair. The writeup proves $\log \mathbb{E}[\#m\text{-matchings}] \leq -\frac{2}{5}T + o(N)$ and concludes $\leq -\frac{1}{5}T$ “for all sufficiently large n ”. With [Lemma 2.3](#) the error term is $(m+u)(s+1)\varepsilon^{1/99} = 80.2T \cdot 101 \cdot \varepsilon^{1/99}$ with $\varepsilon \leq 1/(4n)$, and it is at most the available margin $0.2T$ exactly when $n \geq 10^{455.6}$. The threshold 10^{456} in [Theorem 1.2](#) is where this happens; the referee’s independent estimate was $n \gtrsim 10^{454.6}$ (with the slightly smaller constant $s-1$). Using the exact value -0.7056 of the exponent instead of -0.4 would lower the threshold to about 10^{416} , still far beyond anything that can be written down.

Idea of proof. The expected number of m -matchings in the sampled facet hypergraph is $p^m a_m(F)$. Apply [Lemma 2.3](#) with $x = 1/5$, $D = n$: the main term is $T[x \log(100/x) + x - 100f(x)]$ with $f(x) = x + (1-x) \log(1-x) \geq x^2/2$, hence at most $T[(\log 500 + 1)/5 - 2] < -\frac{2}{5}T$; the error term is at most $8100.2T(4n)^{-1/99} \leq 0.198T$ for $n \geq 10^{456}$. Markov’s inequality finishes. [\[full proof\]](#)

Proposition 3.2 (Pruning). *Delete from H_{rand} every edge containing a 99-set of codegree greater than 200, and let H be the result and $Z = |E(H)|$. Then every 99-set of vertices lies in at most 200 edges of H , $Z \leq 200T$ always, and $\mathbb{E}Z \geq 100T(1 - 100(e/4)^{100}) > 99T$. Consequently, if $n \geq 10^{456}$ is a multiple of 5, some outcome satisfies both A and $Z > 90T$.*

Idea of proof. Conditional on an edge being present, the number of other sampled edges through one of its facets is $\text{Bin}(n-1, 100/n)$, and Markov’s inequality on 2^X gives $\Pr(X \geq 200) \leq 2^{-200}(1 + 100/n)^{n-1} \leq (e/4)^{100} =: q \approx 1.7 \cdot 10^{-17}$; a union bound over the 100 facets gives $\mathbb{E}Z \geq 100T(1 - 100q)$. The deterministic bound $Z \leq 200T$ follows by summing facet degrees. Then $\mathbb{E}[Z\mathbf{1}_A] \geq 99T - 200Te^{-T/5} > 90T$. [\[full proof\]](#)

4 Remarks

Remark 4.1 (The counterexample cannot be exhibited). [Theorem 1.2](#) is an existence statement about an object with $100n \geq 10^{458}$ vertices and about $90n^{99} \approx 10^{45150}$ edges. Nothing of this size can be constructed, stored or spot-checked, and the disproof therefore rests entirely on the correctness of [Lemmas 2.1 to 2.3](#). The referee, unable to test the counterexample itself, tested every ingredient instead: inequality (2) by exhaustive enumeration on 95 weighted hypergraphs with $s \in \{2, 3, 4, 5\}$ and at most 12 vertices (no violation; typical near-tight cases $K_{6,6}$ with $\log Z = 6.579$ against the bound 7.621, and $\text{AG}(2, 3)$ with 1.386 against 2.150); the auxiliary construction of [Lemma 2.3](#) exactly on $C_4, C_6, K_4, K_{3,3}$ and $\text{AG}(2, 3)$; and the constant $-(s-1)$ in (3) independently, as the first moment of the number of m -matchings in a random D -regular s -uniform hypergraph, which agrees with the right-hand side of (3) to within 10^{-4} relative for $s \in \{2, 3, 5, 100\}$. The error term in (3) is genuinely needed: on K_N with $x = 1$ the bound without it fails by an additive 0.847 at every N tested up to 10^8 . This is why the referee’s confidence is medium rather than high, and why a second expert reading of [Lemma 2.1](#) is warranted before any claim of priority.

Remark 4.2 (Sharpness of the conjecture and size of any counterexample). The bound $r + d - 1$ is attained with equality on complete d -uniform hypergraphs for the ten pairs $(d, |V|)$ listed in the introduction, and the referee’s 2357 random instances with $d \in \{3, 4, 5\}$ and $|V| \leq d + 4$ produced no violation, with minimum slack 0. So the conjecture is sharp and non-trivial, no small counterexample exists in that range, and a counterexample had to be a large sparse object of the kind produced here. Nothing in this note concerns $d = 3$ or other small uniformities, or small r ; [Conjecture 1.1](#) remains open there.

Remark 4.3 (Consistency with known theorems; the numbers). The mechanism is a first-moment count showing that a random sparse d -partite hypergraph has no matching covering a fifth of the facets, so that $\chi_f \geq |E(H)|/(T/5) \approx 5d$. It does not conflict with the Pippenger–Spencer and Kahn theorems [3, 4], which require the uniformity k to be fixed while $\Delta \rightarrow \infty$; here $k = d = 100$

and $\Delta \leq 200$. The referee recomputed every constant: the first-moment exponent at $x = 1/5$ is exactly -0.7056 (the writeup’s weakened bound is $-0.557 < -0.4$); $q = (e/4)^{100} = 1.67 \cdot 10^{-17}$; and $450(m-1) = 90T - 450 < 90T$. The method is not pushed to its limit: the first-moment exponent vanishes at $x^* = 0.1435$, so the same argument with $|E(H)| > 99T$ gives $\chi_f \geq 690$, and a random-greedy heuristic (which covers a fraction 0.0887 of the facets) suggests the true value is nearer 1100.

Remark 4.4 (Two points a reader may stumble on). (i) In [Lemma 2.1](#) an edge of M is bad relative to M , since $|e \cap e| = s \geq 2$; this is deliberate and necessary, because $S_v \geq w(M(v))$ forces $\varepsilon \geq w(M(v))/\Delta$, and a reader who excludes $e = f$ obtains a false inequality (a single edge has $Z = 1$ against a “bound” of $e^{-(s-1)}$). (ii) The error term in [Lemma 2.3](#) depends on F only through (N, D, s, x) ; the writeup’s claim that it is uniform over F is thereby justified, although the application does not need it, as F is fixed for each n .

Remark 4.5 (Novelty). The referee found no counterexample to [Conjecture 1.1](#) and no paper attacking it, in the OPG entry (fetched live; the only post-2007 content is a 2009 request for a reference) or in the neighbouring literature cited in the introduction. Novelty was checked only against what could be found online, and the counting lemma is a routine variant of a known entropy argument; what is new, if anything, is its application to Rosenfeld’s problem.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-6-astra (reasoning effort max, single autonomous pass) on 2026-09-16, as part of a sweep over the open problems of the Open Problem Garden (catalog id a_generalization_of_vizings_theorem, leg attacks_opg; original writeup in attacks_opg/a_generalization_of_vizings_theorem/output.md). The model claimed the verdict “disproved” with high confidence. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It re-derived every step, brute-forced inequality (2) on 95 instances and the auxiliary construction on five, confirmed the constant $-(s-1)$ independently, recomputed all arithmetic, tested the conjecture itself on complete and on 2357 random small hypergraphs, fetched the OPG entry and searched for prior art. Its verdict was MINOR_GAPS with confidence *medium*: no step was wrong, but the $o(N)$ term of the counting lemma was never quantified, and its quantification (which the referee carried out) shows that the counterexample exists only for $n \gtrsim 10^{455}$. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised, with full proofs in [Appendix A](#), and the following changes were made, each marked where it occurs:

- [Lemma 2.2](#) (explicit rate $(s+1)\varepsilon^{1/(s-1)}$ for the integral) was added; it replaces the writeup’s unquantified limit (4). The constant $s+1$, the bad-edge bound s^3/b and the threshold 10^{456} are the rewriter’s own explicit choices, made along the lines of the referee’s computation; the referee did not see this exact form.
- [Lemma 2.3](#) is stated with an explicit error term: $\varepsilon = x/D + s^3/b$ (the writeup has $\Delta^{-1} + O_{s,x}(N^{-1})$, with a sketched union bound that is here carried out), and the Stirling step (9) of the writeup is replaced by the one-sided bounds $\log b! \geq b \log b - b$, $\binom{b}{s-1} \leq b^{s-1}/(s-1)!$, which remove its $o(N)$.
- [Proposition 3.1](#) and [Theorem 1.2](#) state the explicit threshold $n \geq 10^{456}$ where the writeup says “sufficiently large n ”; the abstract and [Remark 4.1](#) state the size of the resulting object.
- The two-sentence entropy step (6) of the writeup is written out in the proof of [Lemma 2.1](#), following the referee’s expansion (chain rule conditional on the labels, “ v uncovered” = “ v has the largest label in $M(v)$ ”, each edge charged once, Gibbs’ inequality).
- The writeup’s §4 is stated as [Theorem 1.4](#) with the order of limits made explicit (fix $d \geq 15$, then $n \rightarrow \infty$) and the constant $1 - 2d(e/4)^d$ in place of $1 - o_d(1)$; the writeup mixed $o_d(1)$ and

$o_n(1)$.

- The observation that every edge of M is bad relative to M is stated in [Lemma 2.1](#) and explained in [Remark 4.4](#), as the referee recommended.
- The introduction, the citations [\[2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\]](#), the facet-hypergraph reformulation with the projective-plane example, and the contents of [Section 4](#) are taken from the referee report; the writeup cites nothing and calls its counting lemma new. The entropy method is attributed to Radhakrishnan and to the proofs of the Kahn–Lovász theorem.

No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate this proof for human review, not as a claim to a new resolution.

A Full proofs

Lemma 2.1 (Weighted perfect matchings). *Let $s \geq 2$ and let K be an s -uniform hypergraph on L vertices with positive edge weights w , such that every vertex has weighted degree $\sum_{e \ni v} w(e) \leq \Delta$. Put $Z = \sum_M \prod_{e \in M} w(e)$, the sum over perfect matchings M of K . Call an edge e bad relative to M if $|e \cap f| \geq 2$ for some $f \in M$; note that every edge of M is bad relative to M . Suppose that for every perfect matching M and every vertex v ,*

$$\sum_{e \ni v, e \text{ bad relative to } M} w(e) \leq \varepsilon \Delta. \quad (1)$$

Then

$$\log Z \leq \frac{L}{s} \left[\log \Delta + \int_0^1 \log(u^{s-1} + \varepsilon) du \right]. \quad (2)$$

Proof. If $Z = 0$ there is nothing to prove, so assume K has a perfect matching. Let M be a random perfect matching with $\Pr(M) = \prod_{e \in M} w(e)/Z$. Taking expectations in $-\log \Pr(M) = \log Z - \sum_{e \in M} \log w(e)$ gives

$$H(M) + \mathbb{E} \sum_{e \in M} \log w(e) = \log Z. \quad (5)$$

The reveal process. Independently of M , give each vertex y a label ℓ_y , the ℓ_y being i.i.d. uniform on $[0, 1]$; almost surely they are distinct. For a vertex v write $M(v)$ for the edge of M containing v , and say that v is *first in* $M(v)$ if $\ell_v = \max_{y \in M(v)} \ell_y$. Given M and the labels, let $R_v = \{M(y) : \ell_y > \ell_v\}$ be the set of matching edges containing a vertex of larger label than v , and let

$$S_v = \sum \{w(e) : e \ni v, e \cap f = \emptyset \text{ for all } f \in R_v\}$$

be the total weight of the edges at v that avoid all edges of R_v . If v is first in $M(v)$ then $M(v) \notin R_v$ and $M(v)$ is disjoint from every edge of R_v , so $S_v \geq w(M(v)) > 0$.

Chain rule. Since the labels ℓ are independent of M , $H(M) = H(M \mid \ell)$. Given ℓ , list the vertices $v_1, v_2, \dots, v_L$ in decreasing order of label. The matching M is determined by the sequence $(M(v_1), \dots, M(v_L))$, so by the chain rule

$$H(M \mid \ell) = \sum_{i=1}^L H(M(v_i) \mid \ell, M(v_1), \dots, M(v_{i-1})).$$

Fix i and a value of the conditioning, i.e. the labels and the edges $M(v_1), \dots, M(v_{i-1})$; these edges are exactly the elements of R_{v_i} . If v_i lies in one of them, then v_i is not first in $M(v_i)$ and $M(v_i)$ is determined, so the i th term vanishes. Otherwise v_i is first in $M(v_i)$, and since M is a matching, the conditional distribution P of $M(v_i)$ is supported on the set $\mathcal{A}$ of edges $e \ni v_i$ disjoint from all edges of R_{v_i} , whose total weight is S_{v_i} . Gibbs' inequality, in the form

$$H(P) + \sum_{e \in \mathcal{A}} P(e) \log w(e) \leq \log \sum_{e \in \mathcal{A}} w(e)$$

(the difference of the two sides is $-D(P\|Q) \leq 0$ for $Q(e) = w(e)/\sum_{\mathcal{A}} w$), gives

$$H(M(v_i) \mid \text{conditioning}) + \mathbb{E}[\log w(M(v_i)) \mid \text{conditioning}] \leq \log S_{v_i}$$

whenever v_i is first in $M(v_i)$. Averaging over the conditioning and summing over i ,

$$H(M) + \mathbb{E} \sum_v \mathbf{1}\{v \text{ first in } M(v)\} \log w(M(v)) \leq \sum_v \mathbb{E}[\mathbf{1}\{v \text{ first in } M(v)\} \log S_v].$$

Every edge $f \in M$ has exactly one vertex that is first in f , so the sum on the left equals $\sum_{f \in M} \log w(f)$, and by (5)

$$\log Z \leq \sum_v \mathbb{E}[\mathbf{1}\{v \text{ first in } M(v)\} \log S_v]. \quad (6)$$

Bounding one term. Fix a vertex v and a perfect matching M , and condition on M and on $\ell_v = t$. The other $s-1$ vertices of $M(v)$ have independent labels, so $\Pr(v \text{ first in } M(v) \mid M, t) = t^{s-1}$. Condition further on this event, which involves only the labels of the vertices of $M(v)$. Let $e \ni v$ be an edge that is not bad relative to M . Then $|e \cap f| \leq 1$ for all $f \in M$; in particular $e \cap M(v) = \{v\}$, and the other $s-1$ vertices y of e lie in $s-1$ pairwise distinct matching edges $M(y)$, all different from $M(v)$. The edge e contributes to S_v iff none of these $M(y)$ belongs to R_v , i.e. iff all $s(s-1)$ vertices of $\bigcup_{y \in e \setminus \{v\}} M(y)$ have labels below t . These vertices are disjoint from $M(v)$, so their labels are independent of the conditioning, and the probability is $t^{s(s-1)}$. The bad edges at v have total weight at most $\varepsilon \Delta$ by (1). Hence

$$\mathbb{E}[S_v \mid M, \ell_v = t, v \text{ first in } M(v)] \leq \Delta t^{s(s-1)} + \varepsilon \Delta.$$

Since $S_v > 0$ on the event, Jensen's inequality for the concave function $\log$ gives

$$\mathbb{E}[\mathbf{1}\{v \text{ first in } M(v)\} \log S_v \mid M] = \int_0^1 t^{s-1} \mathbb{E}[\log S_v \mid M, t, \text{first}] dt \leq \int_0^1 t^{s-1} \log(\Delta(t^{s(s-1)} + \varepsilon)) dt.$$

This bound does not depend on M or v . Summing over the L vertices in (6) and substituting $u = t^s$ (so $du = s t^{s-1} dt$ and $t^{s(s-1)} = u^{s-1}$),

$$\log Z \leq L \int_0^1 t^{s-1} \log(\Delta(t^{s(s-1)} + \varepsilon)) dt = \frac{L}{s} \int_0^1 \log(\Delta(u^{s-1} + \varepsilon)) du = \frac{L}{s} \left[\log \Delta + \int_0^1 \log(u^{s-1} + \varepsilon) du \right]. \quad \square$$

Lemma 2.2 (Quantitative form of the integral). *For every integer $s \geq 3$ and every $\varepsilon \in (0, 1)$,*

$$\int_0^1 \log(u^{s-1} + \varepsilon) du \leq -(s-1) + (s+1) \varepsilon^{1/(s-1)}.$$

Proof. Put $u_0 = \varepsilon^{1/(s-1)} \in (0, 1)$, so that $u_0^{s-1} = \varepsilon$ and $(s-1) \log u_0 = \log \varepsilon$.

For $0 \leq u \leq u_0$ we have $u^{s-1} + \varepsilon \leq 2\varepsilon$, hence

$$\int_0^{u_0} \log(u^{s-1} + \varepsilon) du \leq u_0 \log(2\varepsilon) = u_0 \log 2 + u_0 \log \varepsilon.$$

For $u_0 \leq u \leq 1$, write $\log(u^{s-1} + \varepsilon) = (s-1) \log u + \log(1 + \varepsilon u^{-(s-1)}) \leq (s-1) \log u + \varepsilon u^{-(s-1)}$. Now

$$\begin{aligned} \int_{u_0}^1 (s-1) \log u du &= (s-1) [u \log u - u]_{u_0}^1 \\ &= -(s-1) + (s-1)u_0 - (s-1)u_0 \log u_0 \\ &= -(s-1) + (s-1)u_0 - u_0 \log \varepsilon, \end{aligned}$$

and, as $s \geq 3$,

$$\begin{aligned} \int_{u_0}^1 \varepsilon u^{-(s-1)} du &= \frac{\varepsilon(u_0^{-(s-2)} - 1)}{s-2} \leq \frac{\varepsilon u_0^{-(s-2)}}{s-2} \\ &= \frac{u_0^{s-1-(s-2)}}{s-2} = \frac{u_0}{s-2}. \end{aligned}$$

Adding the three estimates, the terms $\pm u_0 \log \varepsilon$ cancel and

$$\begin{aligned} \int_0^1 \log(u^{s-1} + \varepsilon) du &\leq -(s-1) + u_0 \left[(s-1) + \log 2 + \frac{1}{s-2} \right] \\ &\leq -(s-1) + (s+1)u_0, \end{aligned}$$

since $\log 2 + 1/(s-2) \leq \log 2 + 1 < 2$ for $s \geq 3$. $\square$

Lemma 2.3 (Matchings in a regular linear hypergraph). *Let $s \geq 3$ and $x \in (0, 1)$. Let F be an s -uniform, D -regular, linear hypergraph on N vertices such that $m = xN/s$ is an integer, and put $u = N - sm = (1-x)N$ and $b = (s-1)u$. Assume $b \geq 4$ and*

$$\varepsilon := \frac{x}{D} + \frac{s^3}{b} < 1.$$

Then the number $a_m(F)$ of matchings of size m in F satisfies

$$\log a_m(F) \leq m(\log(D/x) - (s-1)) - N(1-x)\log(1-x) + (m+u)(s+1)\varepsilon^{1/(s-1)}. \quad (3)$$

Proof. Recall $u = N - sm = (1-x)N$ and $b = (s-1)u$.

The auxiliary hypergraph. Adjoin to $V(F)$ a set B of b new vertices, and let K be the weighted s -uniform hypergraph on $V(F) \cup B$ ($L = N + b = s(m+u)$ vertices) with the edges of F , each of weight 1, and for every old vertex $a \in V(F)$ and every $C \in \binom{B}{s-1}$ the auxiliary edge $\{a\} \cup C$ of weight

$$w = \frac{(1-x)D/x}{\binom{b}{s-1}}.$$

An old vertex has weighted degree $D + \binom{b}{s-1}w = D + (1-x)D/x = D/x$. A new vertex lies in $N\binom{b-1}{s-2}$ auxiliary edges, and $\binom{b-1}{s-2}/\binom{b}{s-1} = (s-1)/b$, so its weighted degree is $N\frac{s-1}{b} \cdot \frac{(1-x)D}{x} = \frac{(s-1)(1-x)ND}{(s-1)(1-x)Nx} = D/x$. Thus K is weighted-regular with $\Delta = D/x$.

Perfect matchings of K . Let M be a perfect matching of K . Each auxiliary edge contains exactly $s-1$ new vertices and each original edge none, so the auxiliary edges of M partition B and there are exactly $b/(s-1) = u$ of them; the remaining $(N-u)/s = m$ edges of M are original and form an m -matching of F , whose u uncovered old vertices are exactly the old vertices of the auxiliary edges. Conversely, every m -matching of F extends to a perfect matching of K by choosing an ordered partition of B into u blocks of size $s-1$, one for each uncovered old vertex, in $b!/((s-1)!)^u$ ways. Hence

$$Z = a_m(F) \cdot \frac{b!}{((s-1)!)^u} \cdot w^u,$$

which is (4).

Hypothesis (1). Fix a perfect matching M and a vertex v . An edge e is bad relative to M iff two distinct vertices of e lie in a common edge of M .

Original edges at v (so v is old): an auxiliary edge of M contains only one old vertex and so meets an original edge in at most one vertex; an original edge of M meets a different original edge in at most

one vertex by linearity of F . So the only original edge at v that can be bad is $M(v)$ itself, of weight 1. The bad original weight at v is at most 1.

Auxiliary edges at v : they all have the same weight w , so their total bad weight is (weighted auxiliary degree of v) $\times$ (fraction of them that are bad) $\leq \Delta \cdot \beta$, where β is the probability that a uniformly random auxiliary edge at v is bad. We bound β by a union bound over the $\binom{s}{2}$ pairs of positions in the edge. Note that an edge of M containing a new vertex is auxiliary and so contains exactly $s - 1$ new vertices and one old vertex.

Case v old. The random edge is $\{v\} \cup C$ with C uniform in $\binom{B}{s-1}$; list C in uniformly random order as $c_1, \dots, c_{s-1}$, so that each c_j is uniform on B and each pair (c_i, c_j) , $i \neq j$, is a uniformly random ordered pair of distinct elements of B . For a pair (v, c_j) : $\Pr(c_j \in M(v)) = |M(v) \cap B|/b \leq (s-1)/b$. For a pair (c_i, c_j) : $\Pr(c_j \in M(c_i)) = |M(c_i) \cap B \setminus \{c_i\}|/(b-1) = (s-2)/(b-1)$. Hence

$$\beta \leq \frac{(s-1)^2}{b} + \binom{s-1}{2} \frac{s-2}{b-1}.$$

Case v new. The random edge is $\{a\} \cup C$ with a uniform on $V(F)$, $v \in C$, and $C \setminus \{v\}$ uniform in $\binom{B \setminus \{v\}}{s-2}$, independently of a . For a pair (a, c) with $c \in C$: given C , a is uniform on $V(F)$ and $M(c)$ has exactly one old vertex, so $\Pr(a \in M(c)) = 1/N$; there are $s-1$ such pairs. For a pair (v, c) with $c \in C \setminus \{v\}$: c is uniform on $B \setminus \{v\}$, so $\Pr(c \in M(v)) = (s-2)/(b-1)$; there are $s-2$ such pairs. For a pair (c, c') of distinct elements of $C \setminus \{v\}$: it is a uniformly random ordered pair of distinct elements of $B \setminus \{v\}$, so $\Pr(c' \in M(c)) \leq (s-2)/(b-2)$; there are $\binom{s-2}{2}$ such pairs. Since $N = b/((s-1)(1-x)) \geq b/(s-1)$, we get $(s-1)/N \leq (s-1)^2/b$, and

$$\beta \leq \frac{(s-1)^2}{b} + \frac{(s-2)^2}{b-1} + \binom{s-2}{2} \frac{s-2}{b-2} \leq \frac{(s-1)^2}{b} + \binom{s-1}{2} \frac{s-2}{b-2},$$

because $(s-2)^2 + \binom{s-2}{2}(s-2) = (s-2)^2(1 + \frac{s-3}{2}) = \frac{(s-2)^2(s-1)}{2} = \binom{s-1}{2}(s-2)$.

In both cases, using $b \geq 4$ (so $b \leq 2(b-2)$),

$$\beta \leq \frac{(s-1)^2 + \frac{1}{2}(s-1)(s-2)^2}{b-2} \leq \frac{\frac{1}{2}(s-1)s^2}{b-2} \leq \frac{s^3}{2(b-2)} \leq \frac{s^3}{b},$$

where the second inequality is $(s-1) + \frac{1}{2}(s-2)^2 \leq \frac{1}{2}s^2$, i.e. $2s-2 + s^2-4s+4 \leq s^2$, i.e. $2 \leq 2s$. Altogether the bad weight at v is at most $1 + \Delta s^3/b = \Delta(x/D + s^3/b) = \varepsilon \Delta$, which is (1).

Upper bound on $\log Z$. By Lemma 2.1 and then Lemma 2.2 (applicable since $s \geq 3$ and $0 < \varepsilon < 1$), with $L/s = m + u$ and $\Delta = D/x$,

$$\log Z \leq (m+u) \left[\log \frac{D}{x} - (s-1) + (s+1)\varepsilon^{1/(s-1)} \right]. \quad (7)$$

Lower bound on the completion factor. Since $\log b! = \sum_{k=1}^b \log k \geq \int_0^b \log t \, dt = b \log b - b$ and $\binom{b}{s-1} \leq b^{s-1}/(s-1)!$,

$$\begin{aligned} \log \left(\frac{b!}{((s-1)!)^u w^u} \right) &= \log b! - u \log(s-1)! + u \log \frac{(1-x)D}{x} - u \log \binom{b}{s-1} \\ &\geq (b \log b - b) - u \log(s-1)! + u \log \frac{(1-x)D}{x} - u((s-1) \log b - \log(s-1)!) \\ &= -b + u \log \frac{(1-x)D}{x} = u \left[\log \frac{(1-x)D}{x} - (s-1) \right], \end{aligned}$$

using $u(s-1) \log b = b \log b$ and $b = (s-1)u$.

Conclusion. Subtracting the last bound from (7) in (4),

$$\begin{aligned}\log a_m(F) &\leq (m+u) \left[\log \frac{D}{x} - (s-1) \right] - u \left[\log \frac{(1-x)D}{x} - (s-1) \right] + (m+u)(s+1)\varepsilon^{1/(s-1)} \\ &= m \left[\log \frac{D}{x} - (s-1) \right] + u \log \frac{1}{1-x} + (m+u)(s+1)\varepsilon^{1/(s-1)},\end{aligned}$$

and $u \log \frac{1}{1-x} = -N(1-x) \log(1-x)$, which is (3). $\square$

Proposition 3.1 (No large colour class). *Let $m = T/5$ and let A be the event that H_{rand} contains no m edges that are pairwise facet-disjoint. If $n \geq 10^{456}$ then $\Pr(A^c) \leq e^{-T/5}$.*

Proof. Colour classes of H_{rand} correspond to matchings of the random edge-subhypergraph F' of F obtained by retaining each edge independently with probability $p = 100/n$. A fixed m -matching of F survives in F' with probability p^m , so the expected number of m -matchings of F' is $p^m a_m(F)$, and by Markov's inequality $\Pr(A^c) \leq p^m a_m(F)$.

Apply Lemma 2.3 with $s = 100$, $x = 1/5$, $D = n$, $N = 100T$, $m = xN/s = T/5$ (an integer as $5 \mid n$), $u = (1-x)N = 80T$, $b = 99u = 7920T$, $m+u = 80.2T$. Here $\varepsilon = x/D + s^3/b = \frac{1}{5n} + \frac{10^6}{7920n^{99}}$. For $n \geq 2$ the second term is at most $\frac{1}{20n}$ (as $n^{98} \geq 2^{98} > 2600$), so $\varepsilon \leq \frac{1}{4n} < 1$; also $b \geq 4$. Then

$$\begin{aligned}\log(p^m a_m(F)) &\leq m \log p + m(\log(n/x) - 99) - N(1-x) \log(1-x) \\ &\quad + (m+u) \cdot 101 \cdot \varepsilon^{1/99} \\ &= \frac{N}{100} \left[x \log \frac{100}{x} + x - 100f(x) \right] + 80.2T \cdot 101 \cdot \varepsilon^{1/99},\end{aligned}$$

where $f(x) = x + (1-x) \log(1-x)$. We used $m \log(pn/x) = m \log(100/x)$ and

$$m(-99) - N(1-x) \log(1-x) = \frac{N}{100} [x - 100x - 100(1-x) \log(1-x)] = \frac{N}{100} [x - 100f(x)].$$

Main term. $f(0) = f'(0) = 0$ and $f''(x) = 1/(1-x) \geq 1$, so $f(x) \geq x^2/2$ and $100f(1/5) \geq 2$. With $\log 500 < 7$,

$$x \log \frac{100}{x} + x - 100f(x) \leq \frac{\log 500 + 1}{5} - 2 < \frac{8}{5} - 2 = -\frac{2}{5},$$

so the main term is at most $-\frac{2}{5} \cdot \frac{N}{100} = -\frac{2}{5}T$. (The exact value of the bracket is -0.7056 .)

Error term. For $n \geq 10^{456}$, $\varepsilon^{1/99} \leq (4n)^{-1/99} \leq (4 \cdot 10^{456})^{-1/99} = 10^{-(456 + \log_{10} 4)/99} < 10^{-4.612} < 2.45 \cdot 10^{-5}$, so the error term is at most $80.2 \cdot 101 \cdot 2.45 \cdot 10^{-5} T < 0.199 T \leq \frac{1}{5}T$.

Hence $\log \mathbb{E}[\#m\text{-matchings of } F'] \leq -\frac{2}{5}T + \frac{1}{5}T = -\frac{T}{5}$, and $\Pr(A^c) \leq e^{-T/5}$. $\square$

Proposition 3.2 (Pruning). *Delete from H_{rand} every edge containing a 99-set of codegree greater than 200, and let H be the result and $Z = |E(H)|$. Then every 99-set of vertices lies in at most 200 edges of H , $Z \leq 200T$ always, and $\mathbb{E}Z \geq 100T(1 - 100(e/4)^{100}) > 99T$. Consequently, if $n \geq 10^{456}$ is a multiple of 5, some outcome satisfies both A and $Z > 90T$.*

Proof. Every 99-set of $V(H)$ that is not a transversal of 99 classes lies in no edge of H_0 ; every transversal 99-set has codegree at most 200 in H by construction. So the codegree condition holds for all 99-sets.

Expected size. Fix an edge e of H_0 and one of its facets f . The facet f lies in exactly n edges of H_0 , namely e and $n-1$ others, each retained independently with probability $100/n$. Conditional on e being retained, the number X of other retained edges through f is $\text{Bin}(n-1, 100/n)$, and f has codegree greater than 200 iff $X \geq 200$. By Markov's inequality applied to 2^X ,

$$\Pr(X \geq 200) \leq 2^{-200} \mathbb{E}2^X = 2^{-200} \left(1 + \frac{100}{n}\right)^{n-1} \leq \frac{e^{100}}{2^{200}} = \left(\frac{e}{4}\right)^{100} =: q,$$

and $q < 10^{-16}$. A union bound over the 100 facets of e shows that e survives in H with probability at least $p(1 - 100q)$, so

$$\mathbb{E}Z \geq n^{100} \cdot \frac{100}{n} (1 - 100q) = 100T(1 - 100q) > 99T.$$

Deterministic bound. Summing codegrees over the $100T$ transversal 99-sets counts every edge of H exactly 100 times, so $100Z \leq 200 \cdot 100T$ and $Z \leq 200T$.

A good outcome. Let $n \geq 10^{456}$ be a multiple of 5. By [Proposition 3.1](#) and the two bounds just proved,

$$\mathbb{E}[Z\mathbf{1}_A] \geq \mathbb{E}Z - 200T\Pr(A^c) > 99T - 200Te^{-T/5} > 90T,$$

the last step because $T = n^{99} \geq 16 > 5 \log(200/9) \approx 15.5$. If every outcome in A had $Z \leq 90T$ then $\mathbb{E}[Z\mathbf{1}_A] \leq 90T$; so some outcome satisfies A and $Z > 90T$. $\square$

Theorem 1.2 (Main theorem). *Let $n \geq 10^{456}$ be a multiple of 5. There exists a simple, 100-partite, 100-uniform hypergraph H with $100n$ vertices such that*

1. *every 99-set of vertices is contained in at most 200 edges, and*
2. *every facet colouring of H uses at least 451 colours.*

Proof. Let $n \geq 10^{456}$ be a multiple of 5 and fix an outcome satisfying A and $|E(H)| > 90T$, which exists by [Proposition 3.2](#). The hypergraph H is a set of distinct transversal 100-subsets of $V_1 \cup \dots \cup V_{100}$, so it is simple, 100-partite and 100-uniform on $100n$ vertices, and by [Proposition 3.2](#) every 99-set lies in at most 200 of its edges; this is (1).

For (2), let c be a facet colouring of H . Each colour class is a family of pairwise facet-disjoint edges of $H \subseteq H_{\text{rand}}$. Since A holds, H_{rand} has no $m = T/5$ pairwise facet-disjoint edges, and deleting edges cannot create such a family, so every colour class has at most $m - 1$ edges. If c used at most 450 colours, then

$$|E(H)| \leq 450(m - 1) < 450 \cdot \frac{T}{5} = 90T,$$

contradicting $|E(H)| > 90T$. Hence c uses at least 451 colours. $\square$

Theorem 1.4 (Large uniformity). *For every integer $d \geq 15$ there exists a simple d -uniform hypergraph H_d in which every $(d - 1)$ -set lies in at most $2d$ edges and*

$$\chi_f(H_d) \geq (1 - 2d(e/4)^d) \frac{d^2}{\lceil 5 \log d \rceil}.$$

Consequently $\chi_f(H_d)/(r + d - 1) \geq (1 - o(1))d/(15 \log d) \rightarrow \infty$ as $d \rightarrow \infty$, where $r = 2d$.

Proof. Fix $d \geq 15$ and put $a = \lceil 5 \log d \rceil$ and $x = a/d$. Since $g(t) = t - 5 \log t$ is increasing for $t \geq 5$ and $g(15) > 1.4$, we have $a \leq 5 \log d + 1 < d$, so $x \in (0, 1)$. Let n be a multiple of d , $T = n^{d-1}$, and repeat the construction of [Section 3](#) with d classes of size n : H_0 is the complete d -partite d -uniform hypergraph, its facet hypergraph F is d -uniform, n -regular and linear on $N = dT$ vertices, each edge is retained with probability $p = d/n$, and edges containing a $(d - 1)$ -set of codegree greater than $2d$ are deleted. Set $m = xN/d = xT = aT/d$, an integer since $d \mid n$ and $d - 1 \geq 1$; $s = d \geq 3$.

First moment. By [Lemma 2.3](#) (for n large enough that $b = (d - 1)(1 - x)N \geq 4$ and $\varepsilon_n = x/n + d^3/b < 1$), exactly as in the proof of [Proposition 3.1](#) with $c = pn = d$,

$$\log \mathbb{E}[\#m\text{-matchings}] \leq T \left[x \log \frac{d}{x} + x - d f(x) \right] + (m + u)(d + 1) \varepsilon_n^{1/(d-1)}.$$

Using $f(x) \geq x^2/2$ and $\log(d/x) = 2 \log d - \log a$, the bracket is at most

$$x \left[\log \frac{d}{x} + 1 - \frac{dx}{2} \right] = x \left[2 \log d - \log a + 1 - \frac{a}{2} \right] \leq x \left[1 - \frac{1}{2} \log d - \log a \right] =: -\gamma_d,$$

where we used $a \geq 5 \log d$; and $\gamma_d > 0$ because $\frac{1}{2} \log d + \log a \geq \frac{1}{2} \log 15 + \log 14 > 1$. Since $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$ with d fixed, there is $n_0(d)$ such that for $n \geq n_0(d)$ the error term is at most $\gamma_d T/2$, whence $\Pr(A^c) \leq e^{-\gamma_d T/2}$, where A is the event that H_{rand} has no m pairwise facet-disjoint edges.

Pruning. As in Proposition 3.2, with $X \sim \text{Bin}(n-1, d/n)$, $\Pr(X \geq 2d) \leq 2^{-2d}(1+d/n)^{n-1} \leq (e/4)^d =: q_d$, so $\mathbb{E}Z \geq dT(1-dq_d)$, and $Z \leq 2dT$ deterministically. For $n \geq n_1(d)$ we have $2e^{-\gamma_d T/2} \leq dq_d$ (as $T = n^{d-1} \rightarrow \infty$), and then

$$\mathbb{E}[Z\mathbf{1}_A] \geq dT(1-dq_d) - 2dT e^{-\gamma_d T/2} \geq dT(1-2dq_d).$$

So for $n \geq \max(n_0(d), n_1(d))$ some outcome satisfies A and $|E(H)| \geq dT(1-2dq_d)$; let H_d be such an H . It is simple, d -uniform, and every $(d-1)$ -set lies in at most $2d$ of its edges.

Colouring. As in the proof of Theorem 1.2, every colour class of a facet colouring of H_d has at most $m-1 < xT$ edges, so

$$\chi_f(H_d) \geq \frac{|E(H_d)|}{m-1} > \frac{dT(1-2dq_d)}{xT} = (1-2d(e/4)^d) \frac{d^2}{a}.$$

Finally, with $r = 2d$, $r + d - 1 = 3d - 1 < 3d$ and $a \leq 5 \log d + 1$, so $\chi_f(H_d)/(r + d - 1) \geq (1 - 2d(e/4)^d) d / (3(5 \log d + 1)) = (1 - o(1)) d / (15 \log d)$, which tends to infinity. The limits were taken in the order “fix d , then $n \rightarrow \infty$ ”; the thresholds $n_0(d), n_1(d)$ are not quantified here. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step and equation numbers refer to that writeup, not to this note (the writeup’s (1), (3), (4), (6), (7), (9) are our (3), (2), Lemma 2.2, (6), (4) and the Stirling step of Lemma 2.3). The scripts it mentions are in `verification_astra/scripts/a_generalization_of_vizings_theorem/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The conjecture (OPG, Rosenfeld 2007, verified against the live page http://www.openproblemgarden.org/op/a_generalization_of_vizings_theorem): for a *simple* d -uniform hypergraph H in which every $(d-1)$ -set lies in at most r edges, there is an $(r+d-1)$ -colouring of $E(H)$ in which two edges sharing $d-1$ vertices get different colours. The writeup calls this a *facet colouring* and attacks it at $d = 100$, $r = 200$ (bound 299).

The formalisation is the intended one:

- “simple” for a d -uniform hypergraph means no repeated edges. The constructed H is a sub-family of the transversal 100-sets of a 100-partite vertex set, so its edges are distinct 100-sets: simple, 100-uniform. It is not “simple” in the (occasional) Berge sense of *linear*, but that reading is untenable here — under it no two edges could share $d-1 \geq 2$ vertices and the conjecture would be vacuous for $d \geq 3$.
- $d = 2$ recovers Vizing, as the OPG discussion demands: “ $d-1 = 1$ set in at most r edges” is $\Delta \leq r$, “share $d-1 = 1$ vertex” is adjacency, bound $r+1$. Correct.
- The 99-sets that are *not* transversals of 99 distinct classes lie in no edge at all, so the codegree hypothesis really does hold for **every** 99-set, not just the transversal ones. The writeup states this explicitly.

I also checked that the conjecture is *sharp*, not a strawman, by computing the exact facet chromatic number of complete d -uniform hypergraphs: for $H = \binom{[d+k]}{d}$ one has $r = k+1$ and the conjectured bound $r+d-1 = d+k$ is attained with equality in many cases ($(d, n) = (2, 3), (2, 5), (3, 4), (3, 5), (3, 6), (4, 5), (5, 6), (5, 7), (6, 7), (7, 8)$). So a counterexample cannot be a small dense object; it must be a large spread-out one, which is what the writeup produces.

A useful equivalent form, which I used throughout as a cross-check: let F be the *facet hypergraph* (vertices = the $(d-1)$ -sets, one edge $\{e\}$ the d facets of e) per $e \in E(H)$). Then F is d -uniform, **linear** (two d -sets sharing two $(d-1)$ -subsets are equal), has $\Delta(F) \leq r$, and a facet colouring of H is exactly a proper *edge* colouring of F . So Rosenfeld’s conjecture says $\chi'(F) \leq \Delta(F) + d - 1$ for every *realisable* F . (For general linear hypergraphs that is wildly false — a projective plane of order q has $k = \Delta = q+1$ and $\chi' = q^2 + q + 1$ — but a projective plane is not realisable as a facet hypergraph, since d -sets pairwise meeting in $d-1$ points form either a star or a subfamily of a $(d+1)$ -set. All of the conjecture’s content is in this realisability constraint, and the writeup respects it by building an actual H .)

A reasonable author of the original problem would accept this as a disproof for $d = 100$ (and, via §4, for all large d), while noting that small uniformities, in particular $d = 3$, remain open. The writeup says exactly that.

B.2 Step-by-step findings

#	Step	Label	Note
1	Lemma (1): $\log a_m(F) \leq m(\log(D/x) - (s-1)) - N(1-x)\log(1-x) + o(N)$	VALID	Normalisation independently confirmed: I computed the first moment of the number of m -matchings in a random D -regular s -uniform hypergraph from exact log-factorials and it equals the RHS to 5–7 decimals for $s \in \{2, 3, 5, 100\}$, $D \in \{50, 1000\}$, $x \in \{0.2, 0.5, 0.9\}$ (ratios 0.99983–0.99999). For $s = 2$, $x = 1$ it is Brégman/Kahn–Lovász.
2	§1.1 setup: Z, Δ , “bad relative to M ”, hypothesis (2)	VALID	Note the definition makes every $e \in M$ bad ($ e \cap e = s \geq 2$). This is not a slip — it is exactly what forces $\varepsilon \geq w(M(v))/\Delta$, and §1.2 accounts for it (“total weight of bad original edges is at most 1”). Without it (3) would be false (a single edge gives $Z = 1$ vs bound $e^{-(s-1)}$).
3	Eq (5): $H(M) + \mathbb{E} \sum_{e \in M} \log w(e) = \log Z$	VALID	Immediate from $\log \Pr(M) = \sum \log w(e) - \log Z$.
4	Eq (6): label process + chain rule	VALID but terse	Verified in full: labels are independent of M , so $H(M) = H(M \mid \text{labels}) \leq \sum_v H(Z_v \mid Z_{>v}, \text{labels})$ with $Z_v = M(v)$ iff v is first in $M(v)$; “ v uncovered” $\Leftrightarrow$ “ v first in $M(v)$ ”; each edge weight is charged once; combining with $H(P) + \sum P_i \log w_i \leq \log \sum w_i$ conditionally gives (6). The writeup compresses this into two sentences.
5	$\Pr(v \text{ first} \mid t) = t^{s-1}$; a non-bad $e \ni v$ survives w.p. $t^{s(s-1)}$; $\mathbb{E}[S_v \mid \cdot] \leq \Delta t^{s(s-1)} + \varepsilon \Delta$	VALID	The $s-1$ other vertices of a non-bad e lie in $s-1$ <i>distinct</i> matching edges, none equal to $M(v)$ (else e would be bad), so the $s(s-1)$ labels are distinct and independent of the conditioning event, which involves only $M(v)$. $t^{s(s-1)} = u^{s-1}$, $t^{s-1} dt = du/s$.
6	Jensen + substitution $u = t^s \Rightarrow$ (3)	VALID	Brute-force verified: 95 explicit weighted hypergraphs ($s = 2, 3, 4, 5$, $L \leq 12$, exact Z , exact optimal ε over all perfect matchings) — no violation; see Computational check.
7	Eq (4): $\varepsilon \rightarrow 0 \Rightarrow \log Z \leq \frac{L}{s}(\log \Delta - (s-1) + o(1))$	GAP	True, but the <i>rate</i> is never given and it is the linchpin. $\int_0^1 \log(u^{s-1} + \varepsilon) du + (s-1) \approx (s-1)\varepsilon^{1/(s-1)}$ (confirmed numerically). For $s = 100$ this is a 99th root: $\varepsilon = 10^{-8}$ still gives an error of 82 per vertex. See Caveats.
8	§1.2 auxiliary K : weights w , weighted (D/x) -regularity	VALID	Re-derived $\binom{b-1}{s-2} / \binom{b}{s-1} = (s-1)/b$, $b = (s-1)N(1-x)$ and verified exactly by code on $C_4, C_6, K_4, K_{3,3}$ and AG(2,3): old and new weighted degrees both exactly D/x .

#	Step	Label	Note
9	(2) holds for K with $\varepsilon = \Delta^{-1} + O(N^{-1})$	VALID	Bad <i>original</i> edges through v : only $M(v)$ itself (linearity; an auxiliary edge has one old vertex). Bad <i>auxiliary</i> edges: $\binom{s}{2}$ pairs, each colliding with prob $O(1/\min\{N, b-s\}) = O(1/N)$; total $\leq \Delta \cdot O(1/N)$. $\Delta/N = n/(x \cdot 100n^{99}) \rightarrow 0$.
10	$\text{PM}(K) \leftrightarrow m$ -matchings of F ; identity (7)	VALID	Verified exactly by code: #PM(K) = $a_m(F) \cdot b!/((s-1)!)^u$ and Z equals (7) on all five test instances; every PM uses exactly u auxiliary edges, as claimed.
11	Stirling step (9)	VALID	Re-derived: the $-u \log((s-1)!)$ from the partition count and the $+u \log((s-1)!)$ inside $\log \binom{b}{s-1}$ cancel, and $b \log b - (s-1)u \log b = 0$ since $b = (s-1)u$; what is left is $u(\log \frac{(1-x)D}{x} - (s-1)) + O(\log N)$. Exactly as printed.
12	(8) – (9) $\Rightarrow$ (1)	VALID	$u[\log(D/x) - \log((1-x)D/x)] = u \log \frac{1}{1-x} = -N(1-x) \log(1-x)$.
13	§2: F is 100-uniform, n -regular, linear; $N = 100T$, $T = n^{99}$	VALID	$100 \cdot n^{99}$ facets; a facet lies in exactly n transversal edges; two transversal 100-sets sharing two 99-subsets coincide.
14	Colour classes $\leftrightarrow$ matchings of F	VALID	Two edges share 99 vertices iff they share a facet.
15	(10): substitute $p = c/D$, $c = pD = 100$	VALID	$\frac{N}{s} [x \log(c/x) + x(1-s) - s(1-x) \log(1-x)]$ agrees with the writeup's
16	$f(x) \geq x^2/2$	VALID	$\frac{N}{s} [x \log(c/x) + x - sf(x)]$. $f'(x) = -\log(1-x)$, $f(0) = f'(0) = 0$, $f'' = 1/(1-x) \geq 1$.
17	Exponent $\leq (\log 500 + 1)/5 - 2 < -2/5$; (12); (13)	VALID	Exact value -0.705594 ; the writeup's own weakened bound is $-0.557078 < -0.4$. First-moment Markov gives $\Pr(A^c) \leq \mathbb{E}[\#]$.
18	§2.2 pruning; $\Pr(X \geq 200) \leq 2^{-200}(1+100/n)^{n-1} \leq (e/4)^{100}$	VALID	$\mathbb{E}2^X = (1+p)^{n-1}$; $q = (e/4)^{100} = 1.67 \times 10^{-17}$, well under the claimed 10^{-10} . Conditional distribution $\text{Bin}(n-1, 100/n)$ and the threshold $(\deg > 200 \Leftrightarrow X \geq 200)$ are both right.
19	$\mathbb{E}Z \geq 100T(1-100q) > 99T$	VALID	$100q = 1.67 \times 10^{-15}$.
20	$Z \leq 200T$ deterministically	VALID	$100Z = \sum_f \deg_H(f) \leq 200 \cdot 100T$.
21	$\mathbb{E}[Z\mathbf{1}_A] > 99T - 200Te^{-T/5} > 90T$; good outcome exists	VALID	If $Z \leq 90T$ on all of A then $\mathbb{E}[Z\mathbf{1}_A] \leq 90T$.
22	§3: $450(m-1) < 90T < E(H) \Rightarrow \chi \geq 451 > 299$	VALID	$450(T/5 - 1) = 90T - 450$.
23	H simple, 100-uniform, max 99-codegree ≤ 200	VALID	Sub-family of distinct transversal 100-sets; pruned facets have degree 0; non-transversal 99-sets have degree 0.

#	Step	Label	Note
24	§4: $x = a/d$, $a = \lceil 5 \log d \rceil$, exponent $x(2 \log d - \log a + 1 - a/2) < 0$; $\chi \geq (1 - o(1))d^2/(5 \log d)$ vs $3d - 1$	VALID (order of limits sloppy)	I recomputed the exponent bound for $d = 20, 50, 100, 1000$: $-2.4124, -1.6687, -1.1923, -0.2184$, all < 0 as claimed. The separation $d^2/(5 \log d) > 3d - 1$ only kicks in around $d \approx 60$ – 100 ($d = 50$: $127.8 < 149$; $d = 100$: $434.3 > 299$), consistent with the writeup’s “for sufficiently large d ”; the ratio does diverge. Only the reading “for each fixed d , let $n \rightarrow \infty$ ” is justified; the writeup mixes $o_d(1)$ and $o_n(1)$ without saying so.
25	“sufficiently large n ” is justified by the proved asymptotic counting lemma”	GAP	Formally true, but the writeup gives the reader no way to see that a valid n exists. Quantifying step 7 (which I did) shows the required n is about 10^{455} .

No step is ERROR. I attacked the two places the task flagged — simplicity and parameter consistency — and both are clean: H is a set of distinct 100-sets, hence simple; and every parameter ($d = 100$, $r = 200$, $c = pD = 100$, $s = 100$, $x = 1/5$, $m = T/5 = xN/s$, $N = 100T$, $T = n^{99}$, $n \equiv 0 \pmod 5$ so $m \in \mathbb{Z}$, $b = (s - 1)u$) is mutually consistent.

B.3 Reference check

The writeup invokes **no** external result (“No unproved conjecture or literature assertion is used”), so there is nothing to misquote; the counting lemma is proved from scratch. I checked:

- **OPG entry** (fetched live, http://www.openproblemgarden.org/op/a_generalization_of_vizings_theorem): statement, the $d = 2$ /Vizing remark and the Kempe-chain comment match `prompt.md` verbatim. The only post-2007 content is a 2009 comment asking for a reference for Rosenfeld’s conjecture; no resolution, no counterexample, no bibliography. Status: open.
- **Priority.** Repeated searches (Rosenfeld + Vizing + hypergraph + codegree; chromatic index of linear k -uniform hypergraphs; $\Delta/\log \Delta$ -type lower bounds) turned up no counterexample to this conjecture and no paper attacking it. The active neighbouring literature is different: Berge–Füredi (arXiv:2403.06850, arXiv:2510.07494), Erdős–Faber–Lovász and Kang–Kelly–Kühn–Methuku–Osthus (arXiv:2110.06181, which concerns *intersecting* edges, not $(d - 1)$ -sharing ones), and Alon–Kim on t -simple hypergraphs. So **not** ALREADY KNOWN as far as I can tell.
- **Consistency with known theorems.** Pippenger–Spencer/Kahn give $\chi'(F) = (1 + o(1))\Delta$ for linear k -uniform F with $\Delta \rightarrow \infty$ and k fixed. The writeup’s regime has $\Delta = 2d$ with $d \rightarrow \infty$, i.e. $k \asymp \Delta$, which those theorems do not cover (and projective planes show they cannot be extended there). No contradiction.

B.4 Computational check

Scripts in `verification_astra/scripts/a_generalization_of_vizings_theorem/`.

1. `check_entropy_bound.py` — brute-force test of inequality (3), which the writeup asserts *exactly* (no error term). For each instance: all perfect matchings enumerated, Z computed exactly, Δ exact, and the *smallest valid* ε computed as $\max_{M,v} \frac{1}{\Delta} \sum_{e \ni v, e \text{ bad rel. } M} w(e)$; RHS by Simpson. 95 instances: $K_{D,D}$ ($D \leq 6$), K_{2k} , complete 3- and 4-uniform hypergraphs, AG(2,3), and 83 random weighted instances ($s = 2, 3, 4, 5$, $L \leq 12$, weights spanning e^{-6} – e^2). **No violation.** Typical near-tight cases: $K_{6,6}$ $\log Z = 6.5793$ vs bound 7.6214; AG(2,3) $\log Z = 1.3863$ vs bound 2.1498.
2. `check_construction_12.py` — §1.2 bookkeeping. On C_4 , C_6 , K_4 , $K_{3,3}$ ($s=2$) and AG(2,3) ($s=3$, $m=2$): weighted degrees of old and new vertices are **exactly** D/x in all five cases; every perfect matching of K uses exactly u auxiliary edges; and identity (7) holds exactly, e.g. AG(2,3): $\#PM(K) = 1080 = a_2(F) \cdot 6!/(2!)^3 = 12 \cdot 90$ and $Z = 2.56 = \text{predicted } 2.56$. (One line prints “FAIL” for C_4 : that is my auxiliary-edge *detector*, which distinguishes aux edges by weight, and for C_4 the aux weight happens to be 1; the counts on that line match.)
3. `check_arithmetic.py` — every number in §§2–4, plus two independent tests of the lemma:
 - exponent at $x = 1/5$, $s = c = 100$: $x \log(c/x) = 1.242922$, $f(x) = 0.02148516 \geq x^2/2 = 0.02$, $sf(x) = 2.148516$, **exponent** $= -0.705594$ (writeup’s weakened value -0.557078 , claim < -0.4 ✓);
 - first-moment threshold $x^* = 0.143523$ — so the writeup’s own method actually yields $\chi \gtrsim 690$ (using $|E(H)| > 99T$), and its claimed 451 is conservative by $\sim 1.5\times$;
 - random-greedy matching heuristic covers 0.0887 of the vertices, comfortably below x^* (a necessary consistency check: had greedy exceeded x^* the lemma would be wrong);
 - $q = (e/4)^{100} = 1.6728 \times 10^{-17}$, $100q = 1.67 \times 10^{-15}$, $\mathbb{E}Z \geq 99.9999 \dots T$;
 - **independent confirmation of the constant** $-(s-1)$: the first moment of m -matchings in a random D -regular s -uniform hypergraph, computed from exact log-factorials, equals the lemma’s RHS to within 10^{-4} relative for $s \in \{2, 3, 5, 100\}$;
 - finite- N test of (1) on K_N ($s = 2$, $D = N - 1$, exact a_m): for $x = 0.2, 0.6$ the bound holds with slack $\rightarrow 0$ per vertex; **for $x = 1$ the bound is violated** by ≈ 0.847 at every N tested ($N = 100, 10^3, 10^4, 10^6, 10^8$), i.e. by an additive $O(1)$, which is $o(N)$ — so (1) is *false without* its $o(N)$ term and is asymptotically exactly tight. This is not an error in the writeup (which keeps the $o(N)$), but it shows the error term is load-bearing.

4. `check_small_cases.py` / `check_small_random.py` — the conjecture itself on small hypergraphs. Complete d -uniform hypergraphs: the bound $r + d - 1$ is attained with equality at $(d, n) = (2, 3), (2, 5), (3, 4), (3, 5), (3, 6), (4, 5), (5, 6), (5, 7), (6, 7), (7, 8)$ (exact χ by branch and bound), and is never exceeded. 2357 random instances with $d \in \{3, 4, 5\}$, $n \leq d + 4$: **0 violations**, minimum slack 0. So there is no small counterexample; the conjecture is sharp and non-trivial.

What I could **not** check computationally: the counterexample itself. It is non-constructive and, by my own quantification of the error term, needs $n \approx 10^{455}$ — about 10^{457} vertices and 10^{45000} edges. Nothing of that kind can be instantiated. I therefore verified every *ingredient* by brute force instead, which is the most that is possible here.

B.5 Caveats

1. **The unquantified $o(N)$ is the whole ball game.** Step (4)'s $o(1)$ is $\int_0^1 \log(u^{s-1} + \varepsilon) du + (s-1) \approx (s-1)\varepsilon^{1/(s-1)}$, with $\varepsilon \approx x/D = x/n$, and it is multiplied by $L/s = (m+u) = 0.802N = 80.2T$. The writeup's margin is only $0.2T$ (it proves $\log \mathbb{E} \leq -0.4T + o(N)$ and needs $\leq -0.2T$). So one needs $99(0.2/n)^{1/99} \cdot 80.2 \cdot 100 < 0.2$, i.e. $n > 10^{454.6}$ (or $10^{436.4}$ using the true exponent -0.7056 instead of the writeup's -0.4). This is finite, so the theorem stands, but the writeup never checks that its $o(N)$ can be beaten by a constant margin — and with a 99th-root decay this is exactly the kind of thing that fails. A referee would require this paragraph. I supplied it; the conclusion survives.
2. **Uniformity claim.** “The error is uniform over F , with s, x fixed” is asserted, never proved. It happens to be true (the error depends only on D, N, s, x through $\varepsilon = x/D + O(1/N)$ and Stirling's $O(\log N)$), and the application does not even need uniformity, since F is fixed for each n .
3. **Order of limits in §4.** $o_d(1)$ and $o_n(1)$ are used interchangeably. Only “for each fixed d , let $n \rightarrow \infty$, then $d \rightarrow \infty$ ” is justified by what is proved.
4. **Compressed entropy step (6).** “applied conditionally at each reveal, followed by the entropy chain rule” hides the argument that “ v uncovered” = “ v first in $M(v)$ ” and that each edge is charged exactly once. Standard (Kahn–Lovász), and correct, but written as one sentence.
5. **Definition of “bad” silently includes $e \in M$.** Correct and necessary, but a careless reader who excludes $e = f$ obtains a false (3); the writeup never flags it.
6. **Edge cases** are fine: $x \in (0, 1)$ and $s \geq 2$ as required; $m = T/5 \in \mathbb{Z}$ because $5 \mid n$; $b = (s-1)u \gg s$; $Z = 0$ handled. The conclusion says nothing about $d = 2, 3$ (where it is Vizing / open), about small r , or about explicit constructions — all stated honestly in the writeup's own “Scope and limitations” and caveats field.
7. **Gap between the claim and what is proved:** none in the restrictive direction — the verdict line (“simple 100-uniform, max 99-codegree ≤ 200 , ≥ 451 colours vs 299”) is exactly what §3 delivers. If anything the writeup undersells: its own first moment gives $x^* = 0.1435$ and hence ≥ 690 colours; and the random-greedy heuristic (covered fraction 0.0887) suggests the truth is nearer 1100.

B.6 Referee summary

I set out to break this and could not. The suspicious-looking parts are all sound: the hypergraph really is simple (a family of distinct transversal 100-sets), the parameters are mutually consistent, the 99-codegree bound holds for *all* 99-sets and not just the transversal ones, and the entropy matching-counting lemma — which is where a single-pass proof would normally invent a false Brégman-type bound — is genuinely correct: I verified its exact form (3) by brute force on 95 weighted hypergraphs, verified the auxiliary-hypergraph bookkeeping of §1.2 exactly on five explicit instances, and independently re-derived its constant $-(s-1)$ as the first moment for random D -regular s -uniform hypergraphs, matching to five digits. All arithmetic in §§2–4 re-derived and confirmed (the exponent is -0.7056 , better than the claimed -0.4 ; $q = 1.67 \times 10^{-17}$; $450(m-1) < 90T$). The one real deficiency is that the proof's single load-bearing approximation, the $o(N)$ in the counting lemma, is never quantified: its true rate is $(s-1)\varepsilon^{1/(s-1)}$, a 99th root for $s = 100$, so the counterexample exists only for $n \gtrsim 10^{455}$, and the writeup gives the reader no way to see that any valid n exists at all. I did that computation and the conclusion survives, so this is MINOR_GAPS rather than MAJOR_GAP, but a journal would demand it before accepting. Confidence is medium rather than high because the result is a non-constructive claim about an object of size 10^{45000} that cannot be instantiated or spot-checked, and because a correct disproof of a 2007 OPG conjecture by a single unrefereed pass is a strong prior against; I found no counterexample to the counterexample, and no prior art (the OPG page is still open with only a 2009 comment asking for a reference), but a second expert reading of §1.1 is warranted before anyone claims priority.

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